Practice Questions for 2024-10-03-DualSimplexMatrix

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1 Questions

1. In Slide 2, the primal problem (P) is initially infeasible. What characteristic of the problem leads to this infeasibility?

   a. The presence of negative coefficients in the objective function.

   b. The presence of equality constraints.

   c. The presence of negative values on the right-hand side of some constraints.

   d. The presence of unbounded variables.

   e. The presence of more variables than constraints.

2. Why is the dual simplex method particularly useful when dealing with linear programs that are initially primal infeasible but dual feasible?

   a. The dual simplex method directly addresses primal infeasibility by iteratively improving the primal solution until feasibility is achieved.

   b. The dual simplex method efficiently handles primal infeasibility by transforming the problem into its dual, which is always feasible.

   c. The dual simplex method leverages the dual feasibility to maintain feasibility throughout the iterations, leading to an optimal solution even if the primal problem starts infeasible.

   d. The dual simplex method is faster than the primal simplex method when dealing with primal infeasible problems.

   e. The dual simplex method is only applicable to problems that are initially primal infeasible.

3. In the context of the dual simplex method, what is the significance of having all negative coefficients in the objective function of the primal problem (as seen in Slide 2)?

   a. It guarantees the existence of a feasible solution for the primal problem.

   b. It ensures that the dual problem is unbounded.

   c. It is a necessary condition for the dual simplex method to start, as it implies dual feasibility at the origin.

   d. It indicates that the primal problem is unbounded.

   e. It is irrelevant to the application of the dual simplex method.

4. How does the dual based Phase I algorithm (Slides 5 and 6) address the situation where both the primal and dual problems are initially infeasible?

    a. It directly modifies the primal constraints to achieve primal feasibility.

    b. It introduces artificial variables into both the primal and dual problems to create an initial feasible solution.

    c. It modifies the objective function of the primal problem to create a dual feasible problem, allowing the dual simplex method to be applied.

    d. It uses a combination of primal and dual simplex methods to iteratively achieve feasibility in both problems.

    e. It proves that if both the primal and dual are infeasible, no solution exists.

5. In Slide 2, the primal problem (P) is initially infeasible. What characteristic of the problem leads to this infeasibility?

    a. The presence of negative coefficients in the objective function.

    b. The presence of negative right-hand side constants in the constraints.

    c. The presence of both equality and inequality constraints.

    d. The presence of unbounded variables.

    e. The presence of redundant constraints.

6. Why is the dual simplex method particularly useful when dealing with linear programs that are initially primal infeasible but dual feasible?

    a. It directly solves the dual problem, which is always feasible.

    b. It maintains primal feasibility throughout the iterations.

    c. It starts with a dual feasible solution and iteratively moves towards primal feasibility while maintaining dual feasibility.

    d. It transforms the primal problem into an equivalent problem that is always primal feasible.

    e. It uses a different algorithm that is not affected by primal infeasibility.

7. In the context of the dual simplex method, what is the significance of having all negative coefficients in the objective function of the primal problem (as seen in Slide 2)?

    a. It guarantees the existence of a feasible solution.

    b. It ensures that the dual problem is unbounded.

    c. It is a necessary condition for the dual problem to be feasible.

    d. It is a sufficient condition for the dual problem to be feasible.

    e. It simplifies the calculations in the dual simplex method.

8. How does the dual based Phase I algorithm (Slides 5 and 6) address the situation where both the primal and dual problems are initially infeasible?

a. It solves the dual problem first to find a feasible solution, then uses that solution to find a feasible primal solution.

b. It modifies the objective function of the primal problem to create a new problem that is dual feasible, then solves this modified problem using the dual simplex method.

c. It introduces artificial variables into both the primal and dual problems to create a feasible starting point.

d. It uses a different algorithm altogether, such as the interior-point method.

e. It determines that the problem is infeasible and terminates.

9. In the context of the dual simplex method, what is the significance of having all negative coefficients in the objective function of the primal problem (as seen in Slide 2)?

   a. It guarantees primal feasibility.

   b. It ensures the dual problem is unbounded.

   c. It is a necessary condition for initiating the dual simplex method, indicating that the primal problem is likely infeasible and the dual problem is feasible.

   d. It implies the primal problem is unbounded.

   e. It is irrelevant to the dual simplex method.

10. In Slide 2, the primal problem (P) is initially infeasible. What is the specific characteristic of the problem that leads to this infeasibility?

    a. The objective function coefficients are all negative.

    b. The constraints are inconsistent.

    c. At least one constraint has a negative right-hand side constant.

    d. The problem is unbounded.

    e. The problem is degenerate.

2 Answers

1. In Slide 2, the primal problem (P) is initially infeasible. What characteristic of the problem leads to this infeasibility?

   a. Answer A is incorrect. While negative coefficients in the objective function are a characteristic of this problem, they do not directly cause infeasibility. The dual simplex method can handle such cases.

   b. Answer B is incorrect. The problem does not contain any equality constraints; all constraints are inequalities.

   c. Answer C is correct. The primal problem is infeasible because two of its constraints have negative right-hand side values (-8 and -7). This means there is no non-negative solution that satisfies all constraints simultaneously.

   d. Answer D is incorrect. The variables are bounded below by zero ($x_1, x_2 \geq 0$), but this does not cause infeasibility.

   e. Answer E is incorrect. The number of variables and constraints does not directly determine infeasibility. Infeasibility arises from the constraints' incompatibility with non-negativity.

2. Why is the dual simplex method particularly useful when dealing with linear programs that are initially primal infeasible but dual feasible?

   a. The dual simplex method doesn't directly improve the primal solution; it works with the dual problem to achieve primal feasibility indirectly.

   b. The dual of a linear program is not always feasible; it depends on the structure of the primal problem. The dual simplex method exploits the situation where the dual is feasible, even if the primal is not.

   c. The dual simplex method starts with a dual feasible solution and iteratively maintains dual feasibility while moving towards primal feasibility. This is in contrast to the primal simplex method, which requires an initial primal feasible solution. If the primal problem is initially infeasible but the dual is feasible, the dual simplex method provides a direct path to a solution.

   d. While the dual simplex method can be efficient in certain cases, it's not inherently faster than the primal simplex method. The efficiency depends on the specific problem structure.

   e. The dual simplex method can be applied to problems that are initially primal feasible as well, although it's particularly advantageous when the primal is infeasible but the dual is feasible.

3. In the context of the dual simplex method, what is the significance of having all negative coefficients in the objective function of the primal problem (as seen in Slide 2)?

   a. Negative coefficients in the objective function do not guarantee primal feasibility; the constraints determine feasibility.

   b. Negative coefficients in the primal objective function do not imply an unbounded dual problem. The boundedness of the dual depends on the constraints.

   c. While not strictly necessary, having all negative coefficients in the objective function of the primal problem often simplifies the initiation of the dual simplex method. It frequently leads to a dual feasible solution at the origin (all slack variables non-negative in the dual dictionary), which is a necessary starting point for the dual simplex algorithm. This is because the negative coefficients in the primal objective function translate to positive constants in the dual objective function.

d. Negative coefficients in the primal objective function do not necessarily mean the primal problem is unbounded. The constraints determine boundedness.

e. The coefficients in the primal objective function are crucial for determining the initial dual feasibility and are therefore relevant to the dual simplex method.

4. How does the dual based Phase I algorithm (Slides 5 and 6) address the situation where both the primal and dual problems are initially infeasible?

   a. The algorithm doesn't directly modify the primal constraints; it modifies the objective function to achieve dual feasibility first.

   b. Artificial variables are not typically used in this specific algorithm. The approach focuses on manipulating the objective function to achieve dual feasibility.

   c. The dual based Phase I algorithm cleverly addresses the infeasibility by focusing on the dual problem. By modifying the objective function of the primal problem (often by changing the signs of the coefficients), it creates a situation where the origin becomes dual feasible. This allows the application of the dual simplex method, which then iteratively works towards primal feasibility and optimality.

   d. While the dual simplex method is used, it's not a combination with the primal simplex method in this specific algorithm. The focus is on achieving dual feasibility first.

   e. The algorithm doesn't prove non-existence; it attempts to find a solution by creating a dual feasible problem first.

5. In Slide 2, the primal problem (P) is initially infeasible. What characteristic of the problem leads to this infeasibility?

   a. Negative coefficients in the objective function do not directly cause primal infeasibility. They affect the direction of improvement but not the feasibility of the solution.

   b. The infeasibility stems from the negative right-hand side constants in two of the constraints. This violates the non-negativity condition for the slack variables, making the problem infeasible in its primal form.

   c. The presence of both equality and inequality constraints does not inherently cause infeasibility. Many linear programs with both types of constraints are feasible.

   d. Unbounded variables, while problematic, do not directly cause primal infeasibility. The problem might be unbounded, but that's a different issue from infeasibility.

   e. Redundant constraints can make a problem less efficient to solve, but they don't necessarily cause infeasibility.

6. Why is the dual simplex method particularly useful when dealing with linear programs that are initially primal infeasible but dual feasible?

   a. The dual simplex method doesn't directly solve the dual problem; it uses the dual information to guide the solution of the primal problem.

   b. The dual simplex method does not maintain primal feasibility throughout the iterations; it starts with a primal infeasible solution and iteratively moves towards primal feasibility.

   c. The dual simplex method leverages the dual feasibility to start the iterative process. It iteratively improves the primal feasibility while maintaining dual feasibility until an optimal solution is found.

d. The dual simplex method doesn't transform the primal problem; it works directly with the primal problem, using the dual information to guide the iterations.

e. The dual simplex method is still a simplex method; it just uses a different pivot rule and starting point compared to the primal simplex method.

7. In the context of the dual simplex method, what is the significance of having all negative coefficients in the objective function of the primal problem (as seen in Slide 2)?

   a. The sign of the objective function coefficients doesn't guarantee feasibility; feasibility depends on the constraints and right-hand side values.

   b. All negative coefficients in the primal objective function do not necessarily imply that the dual problem is unbounded.

   c. It's not a necessary condition; the dual problem can be feasible even if some primal objective coefficients are positive.

   d. While not strictly necessary, having all negative coefficients in the primal objective function is a sufficient condition for the dual problem to be feasible. This is because the dual problem's constraints will be satisfied with $y = 0$.

   e. While it might simplify some aspects, it's not the primary reason for this condition in the dual simplex method.

8. How does the dual based Phase I algorithm (Slides 5 and 6) address the situation where both the primal and dual problems are initially infeasible?

   a. While the dual problem's information is used, the algorithm doesn't solve the dual problem independently first.

   b. The dual-based Phase I algorithm modifies the primal objective function to ensure dual feasibility, allowing the application of the dual simplex method. Once a feasible solution is found for the modified problem, the original problem is addressed.

   c. Artificial variables are not typically introduced in this specific approach; the modification focuses on the objective function.

   d. The dual-based Phase I algorithm remains within the framework of the simplex method.

   e. The algorithm attempts to find a solution; it doesn't prematurely conclude infeasibility.

9. In the context of the dual simplex method, what is the significance of having all negative coefficients in the objective function of the primal problem (as seen in Slide 2)?

   a. Primal feasibility is not guaranteed; in fact, the example in Slide 2 explicitly shows primal infeasibility.

   b. Negative coefficients in the primal objective function do not directly imply that the dual problem is unbounded. The dual problem's feasibility is more relevant.

   c. All negative coefficients in the primal objective function's maximization problem suggest that the primal problem is likely infeasible (due to negative right-hand sides in constraints), while the dual problem is feasible. This setup is a typical starting point for the dual simplex method.

   d. Negative coefficients in the primal objective function do not imply that the primal problem is unbounded. The dual simplex method is used when the primal is infeasible.

e. The sign of the coefficients in the primal objective function is crucial for the dual simplex method's initiation.

10. In Slide 2, the primal problem (P) is initially infeasible. What is the specific characteristic of the problem that leads to this infeasibility?

    a. Answer A is incorrect. While all objective function coefficients being negative is a characteristic often associated with the dual simplex method, it does not directly cause primal infeasibility.

    b. Answer B is incorrect. Inconsistent constraints would mean there is no solution at all, but here, there are solutions; they just don't include the origin.

    c. Answer C is correct. The slide explicitly points out that the constraints $-2x_1 + 4x_2 \leq -8$ and $-x_1 + 3x_2 \leq -7$ have negative right-hand sides, making the primal problem infeasible at the origin ($x_1 = x_2 = 0$).

    d. Answer D is incorrect. Unboundedness refers to the objective function being able to increase infinitely without violating constraints. Infeasibility means there are no feasible solutions at all.

    e. Answer E is incorrect. Degeneracy refers to a situation where a basic feasible solution has more than m variables equal to zero (where m is the number of constraints). This is not directly related to primal infeasibility.